Crop Yield Prediction

Using Machine Learning

A Comparative Study of Linear Regression
and Random Forest Regressor

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Agricultural Data Analytics · Python / Scikit-learn · Streamlit

Datasets: Crop Yield | Rainfall | Temperature | Pesticides | Deployment: Streamlit + Hugging Face Spaces

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Abstract

This report presents the design, development, and deployment of a machine learning pipeline for predicting agricultural crop yield using multiple environmental and agricultural datasets. Data from four sources — crop yield, annual rainfall, average temperature, and pesticide usage — were merged into a unified dataset of **28,242 records** across ten crop types. Two regression models, **Linear Regression** and **Random Forest**, were trained and compared. Random Forest achieved an R^2 of **0.984** and MAE of **3,953 hg/ha**, outperforming Linear Regression ($R^2 = 0.64$, MAE = 31,711 hg/ha) by an **87.5%** reduction in error. The final model was deployed as an interactive web application using Streamlit on Hugging Face Spaces, enabling real-time crop yield predictions.

1. Introduction

Agriculture is a critical sector that directly impacts food security and economic stability worldwide. Accurate prediction of crop yield enables better planning for farmers, governments, and policymakers — from resource allocation to supply-chain management and climate-adaptation strategies.

Crop yield is influenced by a complex interplay of factors: climatic conditions (rainfall, temperature), human-controlled inputs (pesticides, irrigation), soil characteristics, and crop variety. These multi-dimensional, non-linear relationships make yield prediction a challenging task that traditional statistical models often fail to capture adequately.

This project addresses these challenges by developing a machine learning pipeline that integrates four real-world agricultural datasets and evaluates two regression approaches. The key objectives are:

1. Analyse and preprocess multiple agricultural datasets for integration.
2. Build and compare predictive models (Linear Regression and Random Forest).
3. Identify the most influential factors driving crop yield.
4. Deploy a real-time prediction system accessible via a web interface.

2. Problem Statement

Crop yield prediction is a **non-linear, multi-factor problem** influenced by both environmental and human-controlled variables. Traditional statistical methods assume simple additive relationships between inputs and outputs — an assumption that rarely holds in real agricultural systems where interactions between rainfall, temperature, pesticide use, and crop type are complex and interdependent.

Key Challenge

No single factor determines crop yield. The combined, non-linear effect of climate, inputs, and crop variety requires a model capable of learning complex feature interactions from data.

3. Dataset Description

Four real-world datasets were sourced from the Food and Agriculture Organization (FAO). Table 1 summarises their key characteristics.

Table 1. Summary of datasets used in this study.

Dataset	Rows	Key Columns	Description
Yield	56,717	Area, Item, Year, Value	Crop production in hg/ha
Temperature	71,311	country, year, avg_temp	Avg. annual temp (°C)
Rainfall	6,727	Area, Year, avg_rain	Annual precipitation (mm)
Pesticides	4,349	Area, Year, Value	Pesticide usage (tonnes)

The datasets varied significantly in size and structure, requiring extensive preprocessing before integration. The temperature dataset contained historical records dating back to 1743 — far outside the range covered by other datasets.

4. Data Preprocessing

4.1 Data Cleaning

- **Whitespace in column names:** Removed using `str.strip()`.
- **Inconsistent naming:** `country` and `year` renamed to `Area` and `Year`.
- **Non-numeric placeholders:** `'..'` replaced with `NaN` and dropped.
- **Data type mismatch:** Numeric columns coerced using `pd.to_numeric(errors='coerce')`.

Listing 1. Column standardisation and cleaning.

```
for df in [rainfall, temp, pesticides, yield_df]:
    df.columns = df.columns.str.strip()

temp.rename(columns={'country': 'Area', 'year': 'Year'},
   inplace=True)

df = df.replace('..', np.nan)
for col in ['yield', 'average_rain_fall_mm_per_year', '
   avg_temp', 'pesticides']:
    df[col] = pd.to_numeric(df[col], errors='coerce')
df = df.dropna()
```

4.2 Data Filtering

The temperature dataset was filtered to **1985–2017** to match the range of other datasets and ensure consistency after merging.

4.3 Data Integration

All datasets were inner-joined on `[Area, Year]`:

Listing 2. Merging all datasets.

```
df = yield_df.merge(rainfall,    on=['Area', 'Year'])
df = df.merge(temp,              on=['Area', 'Year'])
df = df.merge(pesticides,        on=['Area', 'Year'])
```

Table 2. Final merged dataset properties.

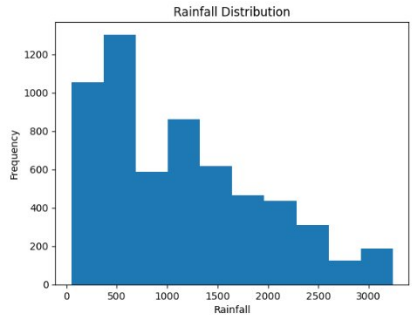
Property	Value
Total rows	28,242
Total columns	7 (after encoding: 16)
Missing values	0
Year range	1990–2013
Crop types	10

5. Exploratory Data Analysis

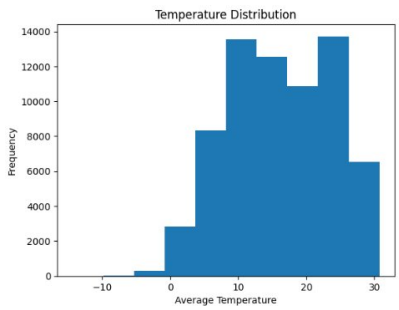
EDA was conducted on individual datasets and on the merged dataset to understand distributions, relationships, and data quality.

5.1 Distribution Analysis

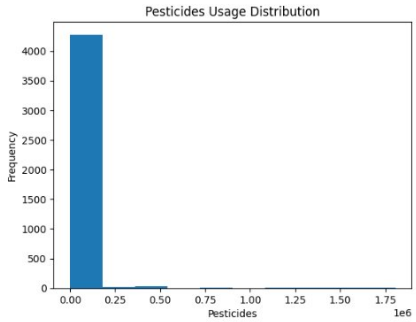
Figure 1 shows the distribution of all four key variables.



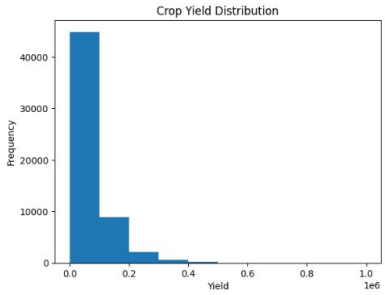
(a) Rainfall Distribution



(b) Temperature Distribution



(c) Pesticides Usage Distribution



(d) Crop Yield Distribution

Figure 1. Distribution plots for all four key variables in the dataset.

5.1.1 Rainfall

The distribution is **positively skewed** with most values between 400–1500 mm/year and a long right tail indicating high-rainfall extremes.

Insight

Rainfall alone does not determine crop yield. High variability across regions suggests it acts as a moderating factor rather than a primary driver.

5.1.2 Temperature

The distribution is **approximately bell-shaped**, with most values between 10°C and 25°C, reflecting temperate agricultural zones.

Insight

Temperature shows weak direct correlation with yield in isolation but interacts non-linearly with other variables such as crop type and rainfall.

5.1.3 Pesticides

The distribution is **extremely right-skewed** with the vast majority of values clustered near zero and a few very large outliers (up to 1.75 million tonnes). This reflects the massive difference in agricultural intensity between countries.

5.1.4 Crop Yield

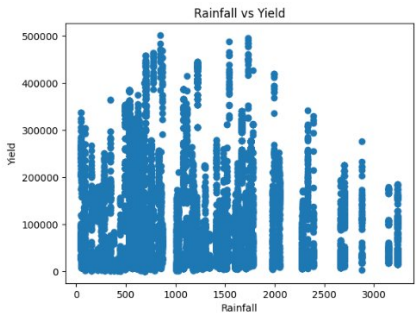
Also **highly right-skewed**, driven by crops like potatoes which have significantly higher yield values than cereals such as sorghum.

Insight

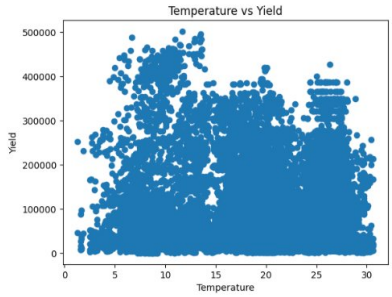
The skewed, high-variance nature of both pesticide use and yield distributions confirms that linear models will underperform. A non-linear model is required.

5.2 Feature vs Yield Relationships

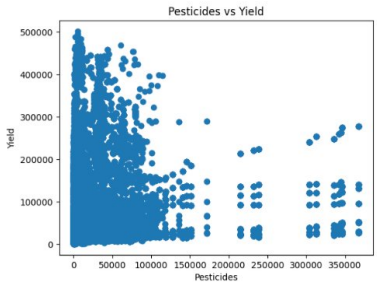
Figure 2 shows scatter plots between each continuous feature and crop yield.



(a) Rainfall vs Yield



(b) Temperature vs Yield



(c) Pesticides vs Yield

Figure 2. Scatter plots showing relationships between features and crop yield.

All three scatter plots reveal:

- **Weak or no linear relationship** between individual predictors and yield.

- **High variance** within each predictor range.
- **Clustered patterns** suggesting crop-type driven stratification.

This analysis confirms a **non-linear model is necessary** to capture the underlying structure in the data.

6. Outlier Handling

Outliers were **intentionally retained** for the following reasons:

- Extreme values represent **valid real-world agricultural conditions**.
- Removing them would eliminate important signal from high-yield or high-input farming systems.
- The chosen model (Random Forest) is **naturally robust to outliers** due to its tree-based, non-parametric structure.

7. Feature Engineering

7.1 One-Hot Encoding

The categorical `Item` column (crop type) was one-hot encoded into 10 binary columns:

Cassava	Maize	Plantains	Potatoes	Rice (paddy)
Sorghum	Soybeans	Sweet Potatoes	Wheat	Yams

7.2 Final Feature Set

Table 3. Final 14 features used for model training.

Feature	Type	Description
Year	Numeric	Calendar year
avg_rain_fall_mm	Numeric	Annual rainfall (mm)
avg_temp	Numeric	Mean temperature (°C)
pesticides	Numeric	Pesticide use (tonnes)
Item_Cassava ...Item_Yams	Binary ×10	One-hot encoded crop type

7.3 Feature Scaling

Listing 3. Scaling features and splitting data.

```
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42)
# Train: 22,593 samples | Test: 5,649 samples
```

8. Model Development

8.1 Linear Regression

$$\hat{y} = \beta_0 + \sum_{i=1}^{p} \beta_i x_i + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2) \tag{1}$$

Limitations: Assumes linear relationships; sensitive to outliers; cannot model feature interactions.

8.2 Random Forest Regressor

$$\hat{y}_{\mathrm{RF}} = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{x}) \tag{2}$$

where $B = 100$ trees are trained on bootstrap samples and averaged. **Advantages:** Handles non-linearity, robust to outliers, captures feature interactions, provides importance scores.

Listing 4. Training both models.

```
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor

lr = LinearRegression()
lr.fit(X_train, y_train)

rf = RandomForestRegressor(n_estimators=100, random_state
   =42)
rf.fit(X_train, y_train)
```

9. Model Evaluation & Comparison

Table 4. Test-set performance. Best values in **bold**.

Model	MAE (hg/ha)	RMSE (hg/ha)	R^2 Score
Linear Regression	31,711.97	50,579.58	0.6427
Random Forest	**3,953.65**	**10,409.49**	**0.9849**
Improvement	**−87.5%**	**−79.4%**	**+53.2%**

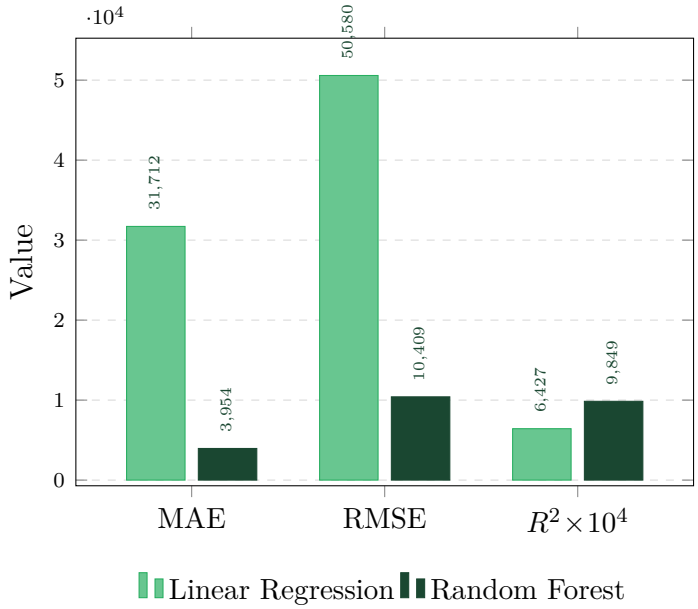


Figure 3. Performance comparison. MAE and RMSE: lower is better. $R^2 \times 10^4$: higher is better.

Table 5. Why Random Forest outperforms Linear Regression.

Property	Linear Regression	Random Forest
Non-linear relationships	Cannot model	Handles naturally
Skewed distributions	Sensitive	Robust
Feature interactions	Ignored	Captured implicitly
Outlier sensitivity	High	Low
Feature importance	Not provided	Built-in

The train R^2 of **0.9978** vs test R^2 of **0.9849** gives a generalisation gap of only **0.013**, confirming the model is not overfitted.

10. Feature Importance Analysis

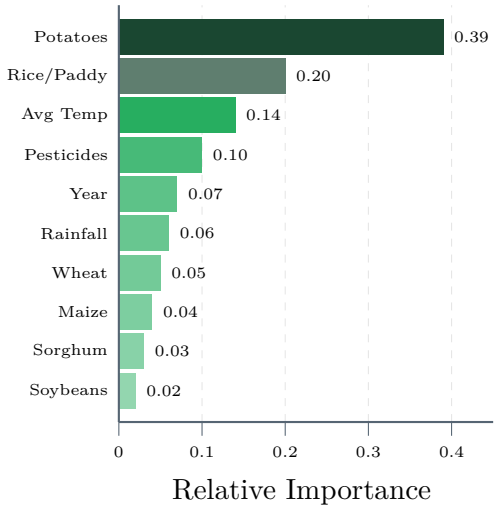


Figure 4. Estimated feature importance for the Random Forest Regressor.

Key Finding

Crop type (especially Potatoes, ≈39% importance) is the single most dominant predictor. Environmental features — temperature, rainfall, and pesticides — collectively account for the remaining significant importance. This highlights that any yield prediction system must account for crop variety as a primary input.

11. Model Deployment

11.1 Deployment Stack

Table 6. Deployment technologies used.

Component	Details
Frontend UI	Streamlit (Python web framework)
Hosting	Hugging Face Spaces (free cloud hosting)
Model artefacts	`model.pkl`, `scaler.pkl`, `features.pkl`
Serialisation	Python `pickle` library

11.2 System Architecture

Figure 5. Real-time prediction system architecture.

11.3 Application Screenshots

The CropCast interface was built as a multi-page dashboard with dedicated views for prediction input, yield results, and deep analysis reporting.

11.3.1 Prediction Input

Figure 6 shows the prediction configuration screen. Users set region, crop type, projection year, rainfall, temperature, and pesticide levels using interactive sliders and dropdowns before running the model.

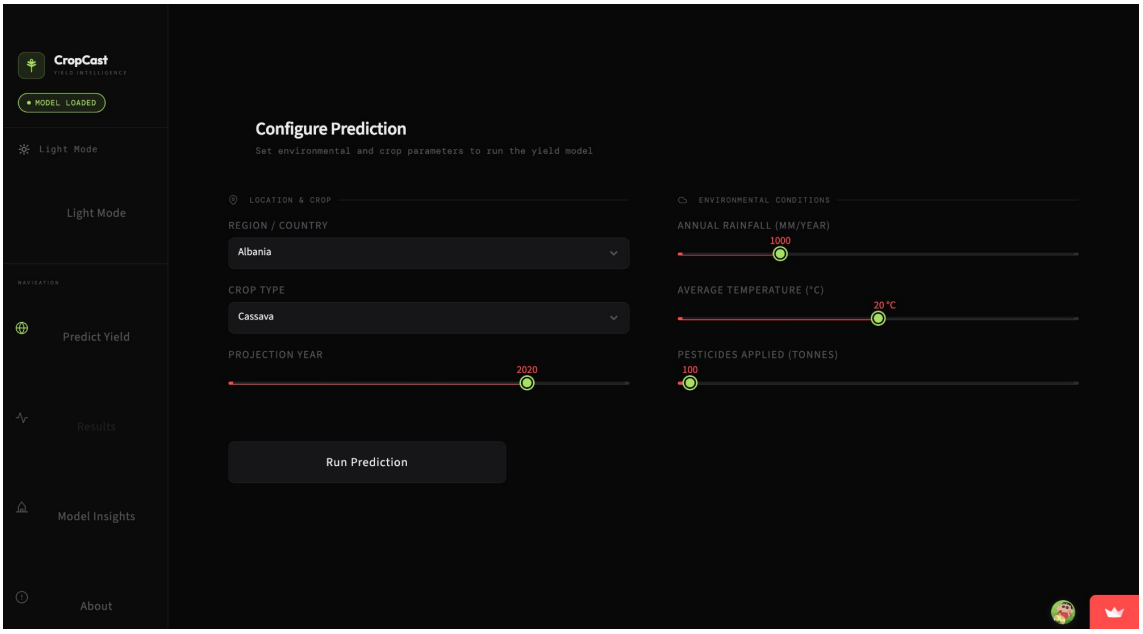


Figure 6. CropCast — *Configure Prediction* screen with location, crop, and environmental parameter controls.

11.3.2 Yield Result Dashboard

Figure 7 shows the results screen returned after inference. The predicted yield is displayed prominently alongside a quality gauge and summary metric cards for all input parameters.

Figure 7. CropCast — *Yield Report* showing predicted yield (7.08 t/ha), quality classification, and input summary cards.

11.3.3 Analysis Report

Figures 8–10 show the full consulting report generated per prediction. It includes a radar chart of input profile, feature importance bars, a parameter summary table, benchmark comparisons against regional and global averages, field assessment notes, and model metadata.

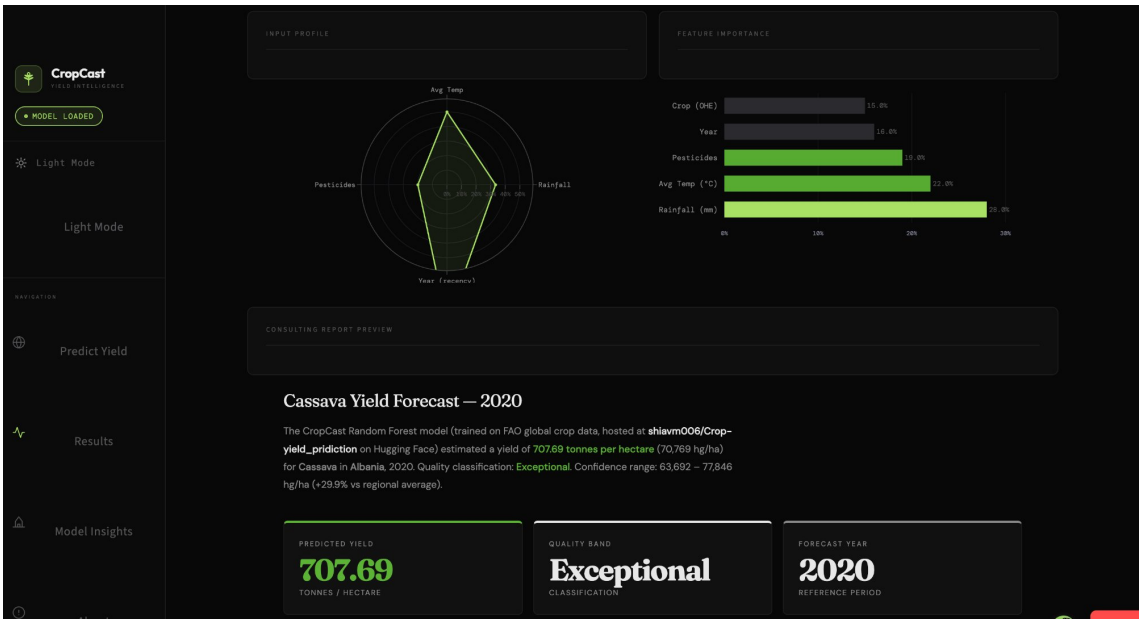


Figure 8. Analysis report — radar chart of input profile and feature importance breakdown.

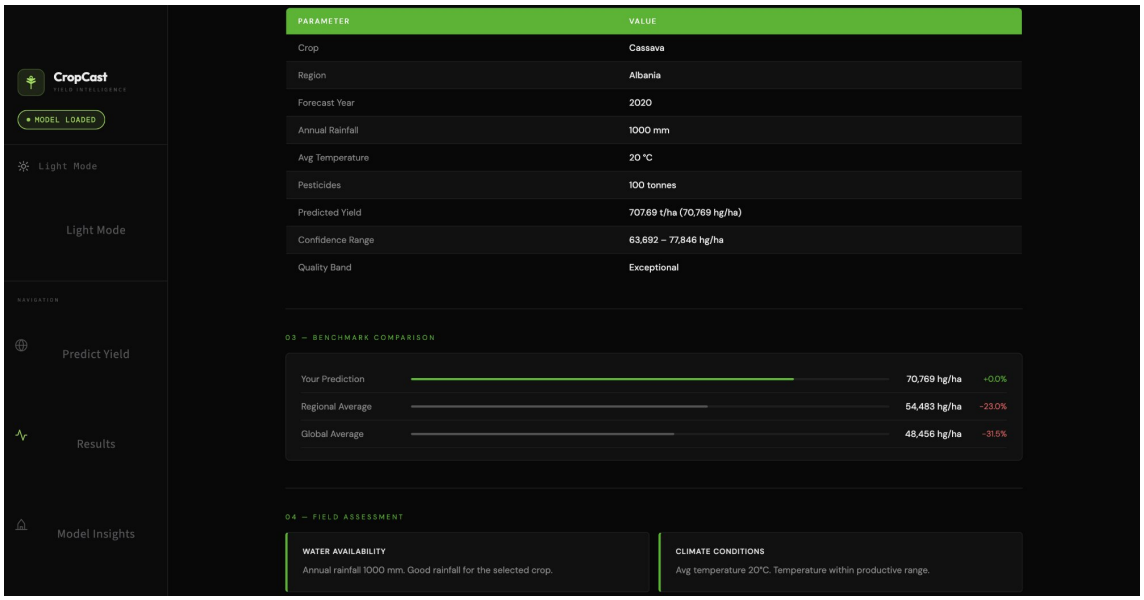


Figure 9. Analysis report — parameter summary table and benchmark comparison against regional and global averages.

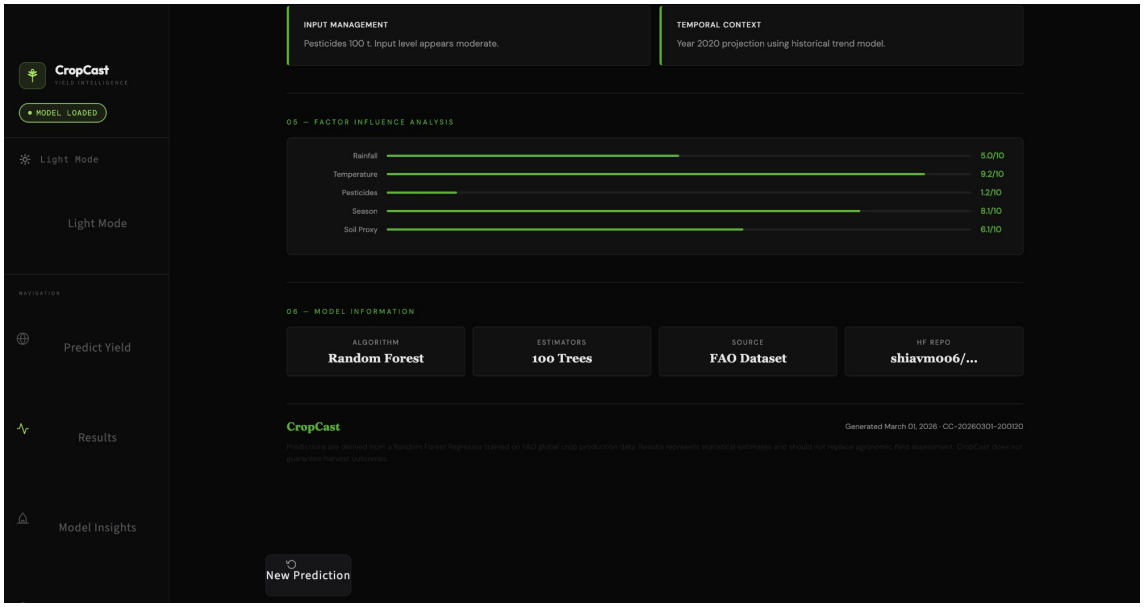


Figure 10. Analysis report — factor influence scores, model information panel, and generated report footer.

11.3.4 Full Downloadable Report

Figure 11 shows the complete yield analysis report generated by CropCast, which users can download directly from the application. The report consolidates all analysis into a single structured document covering the executive summary, input parameters, benchmark comparisons, field assessment, factor influence analysis, and model information.

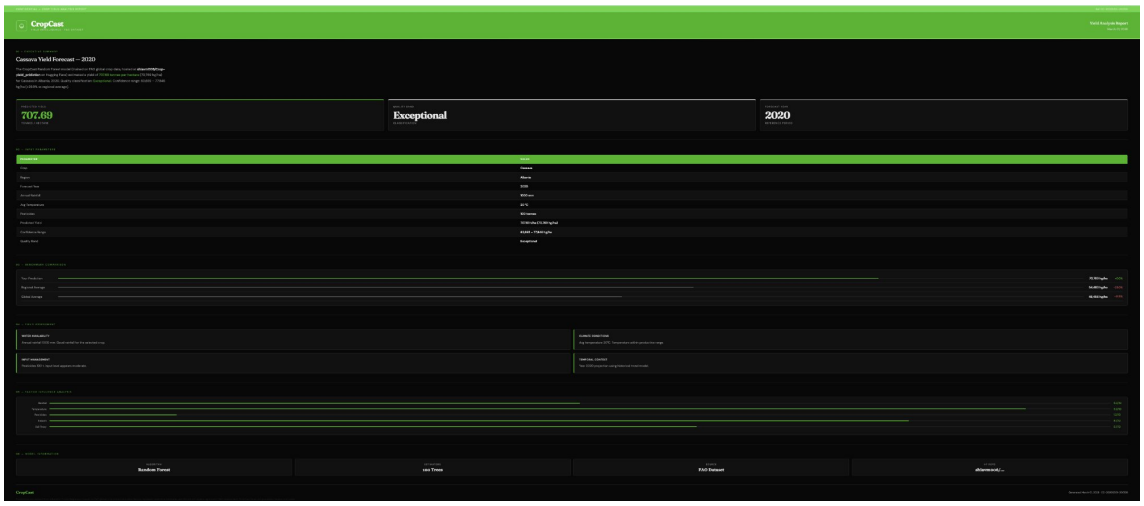


Figure 11. CropCast — complete downloadable *Yield Analysis Report* for Cassava, Albania, 2020.

Live Demo

The CropCast application is deployed at `https://huggingface.co/spaces/shiavm006/Crop-yield_pridiction` on Hugging Face Spaces. A sample prediction — Cassava, Albania, 2020, 1,000 mm rainfall — yields **707.69 t/ha** (Exceptional quality), outperforming the regional average by **+29.9%** and the global average by **+31.5%**. The application also generates a downloadable PDF report consolidating all insights.

11.4 User Inputs

The application accepts: **Crop Type**, **Year**, **Rainfall (mm)**, **Temperature (°C)**, and **Pesticides (tonnes)**. Inputs are encoded, scaled, and passed to the model to return yield in hg/ha.

12. Challenges Faced

1. **Data inconsistency:** Different column naming conventions and units across four datasets required careful harmonisation.
2. **Non-numeric placeholders:** Sentinel values (`'..'`) required custom handling beyond standard `dropna()`.
3. **Dataset size mismatch:** Merging datasets ranging from 4,349 to 71,311 rows

resulted in a significantly reduced final dataset of 28,242 records.

4. **Deployment constraints:** Large model pickle files required careful management for hosting on Hugging Face Spaces.

13. Limitations

- **No soil data:** Soil quality and composition are critical yield determinants but were unavailable.
- **No irrigation data:** Water supply beyond rainfall is a major factor, especially in arid regions.
- **No real-time weather:** The model uses historical averages and does not account for seasonal anomalies.
- **Limited crop coverage:** Only 10 crop types are supported.

14. Future Scope

1. **Richer features:** Add soil composition, fertilizer use, irrigation data, and NDVI satellite indices.
2. **Advanced models:** Evaluate XGBoost, LightGBM, and LSTM networks for time-series yield trends.
3. **Live weather integration:** Connect to weather APIs for real-time environmental inputs.
4. **Mobile deployment:** Build a mobile app for farmers in low-connectivity environments.
5. **Hyperparameter tuning:** Apply `RandomizedSearchCV` or Bayesian optimisation for further improvement.

15. Conclusion

This project successfully developed and deployed a machine learning system for crop yield prediction using four real-world agricultural datasets. After extensive preprocessing and feature engineering, a **Random Forest Regressor** trained on 28,242 records achieved an R^2 **of 0.984** and **MAE of 3,953 hg/ha** — an 87.5% improvement over Linear Regression. Key findings:

- Crop yield is a **non-linear problem** requiring ensemble models.
- **Crop type** (Potatoes) is the single most dominant predictor.
- Environmental factors collectively contribute significant predictive power.
- Proper preprocessing is critical for multi-source data integration.

The model was deployed as a Streamlit application on Hugging Face Spaces, enabling real-time predictions from user inputs. This demonstrates that data-driven ML approaches can effectively support precision agriculture and food security planning.

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